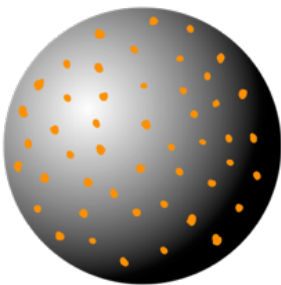


Opener Activity: View Slide

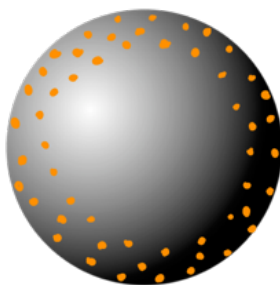
1. A)
B)
C)



2. Two spheres are below with the **same mass and radius**. Based on their mass distributions (orange dots), which is harder to rotate? Why?



Uniform mass
distribution



Hollow mass
distribution

3. Match each statement below to either cortical (C) or trabecular (T) bone.

Found in cuboidal bones, flat bones, and ends of long bones _____

Also called “cancellous” _____

Found in shafts of long bones _____

High porosity _____

Has Haversian and Volkmann’s canals _____

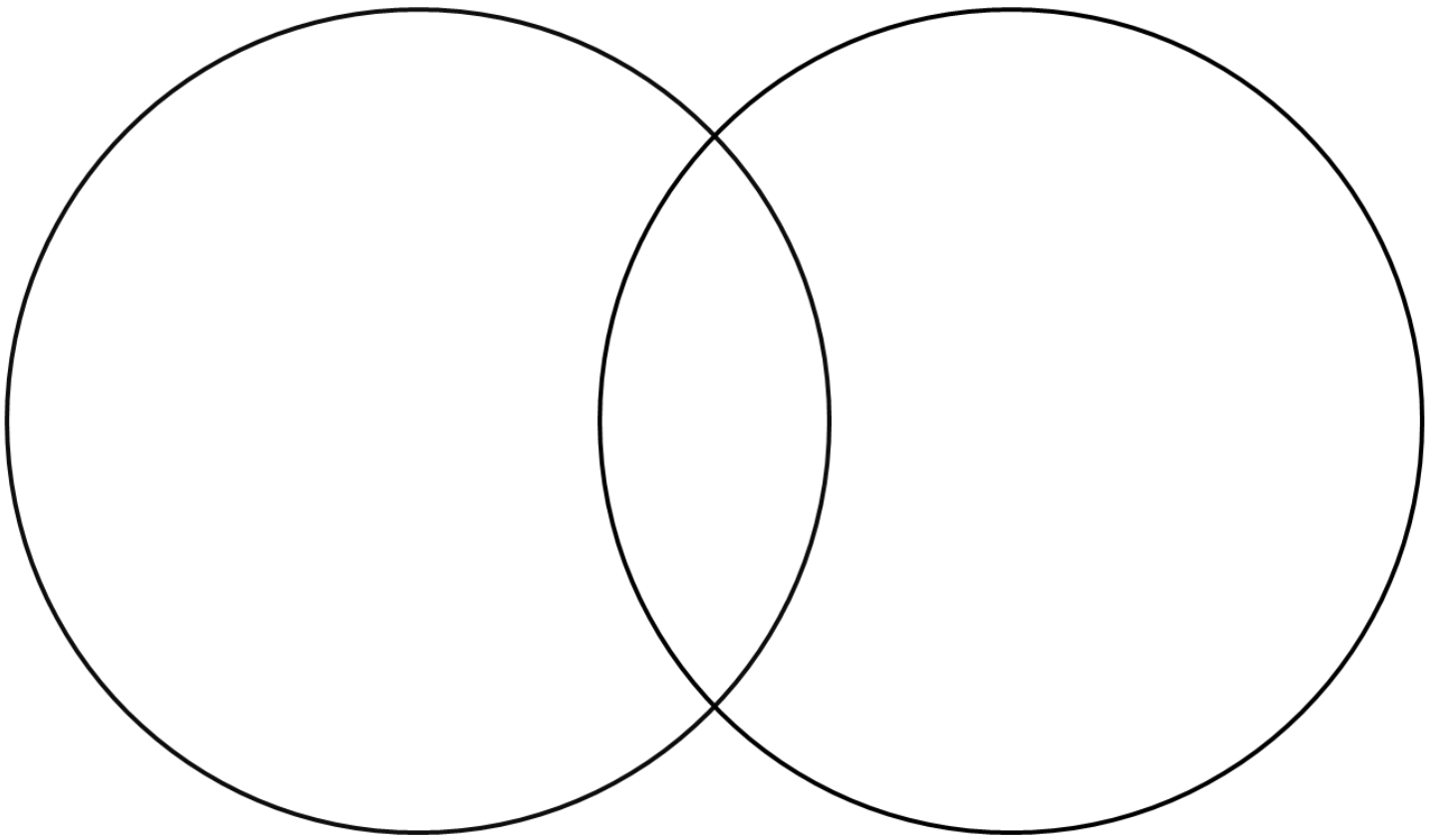
Also called “spongy” _____

Forms the shell around vertebral bodies _____

Also called “compact” _____

Low porosity _____

4. Using the Venn diagram below, compare and contrast lamellar bone and woven bone.



5. **Process Flow Race:**

I am going to separate the class into two or more groups. Each group will receive a set of cutouts representing the process of **bone remodelling**.

Instructions:

1. Each team gets a shuffled set of cutouts.
2. Read through each carefully.
3. Group the cards into phases of bone remodeling that seem to belong together.
4. Once you've made phases, arrange them in the correct chronological order.
5. The first team to correctly group and sequence wins!

Note: Some cutouts are distractors...

Activation	Resorption	Formation
Precursor cell involvement	Osteoclast involvement	Refill gaps in bone
3 days for completion	30 days for completion	3 months for completion
Differentiation	Tunneling through bone	Osteoblast involvement
Increases bone diameter	40 um/day	Osteocyte involvement
	Acid secretion & collagen breakdown	Vascular loop remains
		Haversian canal remains

6. Shariq performs a bicep curl using weighted cables.

- His biceps (B) attach 6cm below the elbow (E), pulling at an angle of 15° clockwise from the forearm.
- His forearm + hand is at an angle of 25° above the horizontal.
- The tension in the cable is 95N, acting at an angle of 63° below the horizontal.
- Assume the forearm + hand is one rigid structure with a length of 45cm.



*Real image of Shariq at the gym

- a) Create a free body diagram **of the forces acting on the right arm.**
- b) Find the bicep force (F_b).
- c) Find the elbow joint force.